Git and GitHub

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What is Git

Git is a software tool that is installed on the desktop/laptop. It is a command-line tool with three versions: GitBash, GitCMD, and Git GUI.

It is designed to manage the various versions of edits (code)made to files with in git repository. Git is a high-quality version control system.

What is GitHub

Git hub is a service which is hosted on the web, it provides a graphical user interface, it is a place where we push/upload the code/copy of the git repository. GitHub is a cloud-based hosting service.

What is version control

Version control is a system which records the changes to files or set of files over a time so that we can recall specific versions later, for ex v1,v2,v3.

Or

Any Application/software or code updated multiple times that is called version control.

Why it requires
Improves the visibility
Helps to team collaborate around the world
Accelerates the product delivery
Allows to merge the changes
Centralize code management
Easy to verify which version is using by team.

Gitrevert is the cmd to go previous version

Source code folder(workspace)
It is a workspace where all code will be saved which may be a java/python application code.

.Git folder

A git software will be tracking and save it memory into this folder.

Think of .git as the **brain/database of your Git repository**.
For example:

```
my-project/
├── index.html
├── app.py
├── README.md
└── .git/
    ├── HEAD
    ├── config
    ├── objects/
    ├── refs/
```

└── index

The .git folder stores information Git needs to track your project.

https://github.com/mahmadkamil/vlc - it is the vlc git hub link

https://github.dev/mahmadkamil/vlc - it will open on vs code directly

https://gitdiagram.com/mahmadkamil/vlc - it will open architecture of this vlc

https://github.com/jubalh/awesome-os - Multiple linux os flavours from linux

```
echo "# Myfirstrepo" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/mahmadkamil/Myfirstrepo.git
git push -u origin main
```

From <https://github.com/mahmadkamil/Myfirstrepo>

```
echo "# Myfirstrepo" >> README.md
```
It will print the string into readme.md file

```
git init
```
It will initialize the empty repository

```
git add README.md
```
Which is add the readme.md file into git and it will be tracked file

```
git commit -m "first kamil commit"
```
We commit it or save it and -m is the message we pass the message is "first kamil commit"

Before commit for the first time we need to configure the account we need to do authorization and authentication

```
git config --global user.mail " mahmad.com"
git config --global user.name "mahmad"
```

```
git branch -M main
```
We change the branch from master to main

Generally our laptop has default master branch GitHub has main branch that's why we changed.

```
git remote add origin https://github.com/mahmadkamil/Myfirstrepo.git
```
connect your local Git repository to your GitHub repository.

```
git push -u origin main
```
We push the code from staging area to GitHub repository

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If you want add some files or you want to changes something into the old repo need to perform below cmds

```
git status
git add file 123.file
git add *
git commit -m "second commit"
git branch -M main
git remote add origin https://github.com/mahmadkamil/Myfirstrepo.git
git push -u origin main
```

git remote add origin https://github.com/mahmadkamil/clock-watch-/tree/Weather-branch

Branch in git
A branch in a git is a simply lightweight movable pointer to one of these commits

What is rebase
It is a way to replay commits one by one on top of a branch

What is git rebase master?
Rebase your current branch on top of the latest commit from master

fast-forward merge
A **fast-forward merge** is when Git can merge one branch into another **without creating a new merge commit**.
Think of it like **moving the branch pointer forward**.

Or

It will just shift the master HEAD.

What is a squash merge?
It will compact or combine all feature commits into one before merging